

Lecture 08

- Exponential matrix
- Linear Systems: Real eigenvalues
- Linear Systems: Complex eigenvalues.

but before

Homework 3.

Q10: $\lambda = 3$, $\vec{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $w = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$

Suggested strategy: linear systems. $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$

2 eq. $(A - 3I) \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

2 eq. $(A - 3I)^2 \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$(A - \lambda I)w = v$

$(A - \lambda I)^2 w = (A - \lambda I)v = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

Possible strategy: Jordan canonical form.

correct. $A = PJP^{-1} = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} \text{jordan block} \\ 3 & 1 \\ & 3 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix}^{-1}$

incorrect: $P \cdot \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \cdot P^{-1} = P \cdot (3I) P^{-1} = 3(P \cdot P^{-1}) = 3I$

MATLAB $[V, D] = \text{eig}(A)$

be careful with $J = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix}$, since

conclusion: on $\mathbb{R}^2$, repeated eigenvalue λ with geometric multiplicity 2 is $A = \lambda \cdot I$.

— end of hw3 q10.

Post class

$$\begin{cases} x' = 1 \\ y' = x \end{cases}$$

Y is sol.

$$Y(t) = \begin{bmatrix} t \\ t^2/2 \end{bmatrix} = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} \quad 2Y(t) = \begin{bmatrix} 2t \\ t^2 \end{bmatrix}$$

$2Y$ is not a sol.

$$x' = Ax, \begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} \neq "x' = Ax"$$

$$x' = Ax + \vec{b}$$

$$Y' = \begin{bmatrix} 1 \\ t \end{bmatrix} = \begin{bmatrix} 1 \\ x \end{bmatrix}$$

$$(2Y)' = 2Y' = \begin{bmatrix} 2 \\ 2t \end{bmatrix} = \begin{bmatrix} 2 \\ x \end{bmatrix} \neq \begin{bmatrix} 1 \\ x \end{bmatrix}$$

$$x' = 1 \Rightarrow x = t + c$$

$$y' = x = t + c \Rightarrow y = \frac{t^2}{2} + ct + d$$

$$\begin{bmatrix} x(t) \\ y(t) \end{bmatrix} = \begin{bmatrix} t + c \\ \frac{t^2}{2} + ct + d \end{bmatrix} \rightarrow \text{all solutions.}$$

$$\left(\frac{t^2}{2} + ct + d \right)' = \frac{2t}{2} + c = t + c = x.$$

— end of post class

Exponential Matrix.

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2} + \frac{x^3}{6} + \frac{x^4}{24} + \dots + \frac{x^n}{n!} + \dots$$

$$e^A = \sum_{n=0}^{\infty} \frac{A^n}{n!} = I + A + \frac{A^2}{2!} + \frac{A^3}{3!} + \dots$$

Example: Compute e^{At} for each A

$$\begin{aligned} \text{a) } A &= \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} & t^2 A^2 &= \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} t^2 = \begin{bmatrix} 2^2 & 0 \\ 0 & 3^2 \end{bmatrix} t^2 \\ & & t^3 A^3 &= \begin{bmatrix} 2^3 & 0 \\ 0 & 3^3 \end{bmatrix} t^3 \end{aligned}$$

$$t^n A^n = \begin{bmatrix} 2^n & 0 \\ 0 & 3^n \end{bmatrix} t^n = \begin{bmatrix} (2t)^n & 0 \\ 0 & (3t)^n \end{bmatrix}$$

$$e^{At} = \sum_{n=0}^{\infty} \begin{bmatrix} (2t)^n & 0 \\ 0 & (3t)^n \end{bmatrix} \cdot \frac{1}{n!} = \begin{bmatrix} \sum \frac{(2t)^n}{n!} & 0 \\ 0 & \sum \frac{(3t)^n}{n!} \end{bmatrix} = \begin{bmatrix} e^{2t} & 0 \\ 0 & e^{3t} \end{bmatrix}$$

$$b) A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$A^3 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$$

$$A^4 = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

$$A^5 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} A = A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$A^k = A^4 \cdot A^2 = I \cdot A^2 = A^2$$

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \frac{t}{1!} + \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \frac{t^2}{2!} + \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \frac{t^3}{3!} + \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \frac{t^4}{4!} + \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \frac{t^5}{5!} + \dots$$

$$\begin{bmatrix} 1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \dots & -t + \frac{t^3}{3!} - \frac{t^5}{5!} + \dots \\ t - \frac{t^3}{3!} + \frac{t^5}{5!} - \dots & 1 - \frac{t^2}{2!} + \frac{t^4}{4!} - \dots \end{bmatrix}$$

$$= \begin{bmatrix} \cos(t) & -\sin(t) \\ \sin(t) & \cos(t) \end{bmatrix} = e^{\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} t}$$

$$c) A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$t^2 A^2 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} t^2 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} t^2 = \begin{bmatrix} t^2 & 2t^2 \\ 0 & t^2 \end{bmatrix}$$

$$t^3 A^3 = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} t^3 & 3t^3 \\ 0 & t^3 \end{bmatrix}$$

$$t^4 A^4 = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 4 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} t^4 & 4t^4 \\ 0 & t^4 \end{bmatrix}$$

$$A^n t^n = \begin{bmatrix} t^n & n t^n \\ 0 & t^n \end{bmatrix}$$

$$\sum_{n=0}^{\infty} \frac{n t^n}{n!} = \sum_{n=0}^{\infty} \frac{t^n}{(n-1)!} = t \left(\sum_{n=0}^{\infty} \frac{t^{n-1}}{(n-1)!} \right) = t \sum_{k=0}^{\infty} \frac{t^k}{k!}$$

$$\sum \frac{A^n t^n}{n!} = \begin{bmatrix} \sum \frac{t^n}{n!} & \sum \frac{n t^n}{n!} \\ 0 & \sum \frac{t^n}{n!} \end{bmatrix} = \begin{bmatrix} e^t & t e^t \\ 0 & e^t \end{bmatrix} = e^{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} t}$$

Natural question:

• Why are doing this? $x' = ax \Rightarrow x(t) = x_0 \cdot e^{at}$

$$x' = Ax \Rightarrow \underbrace{X(t)}_{2 \times 1} = \underbrace{e^{At}}_{2 \times 2} \cdot \underbrace{x_0}_{2 \times 1} = \underbrace{\begin{bmatrix} x(t) \\ y(t) \end{bmatrix}}_{2 \times 1}$$

Example: Verify that $e^{\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} t} \cdot x_0$ is a solution of

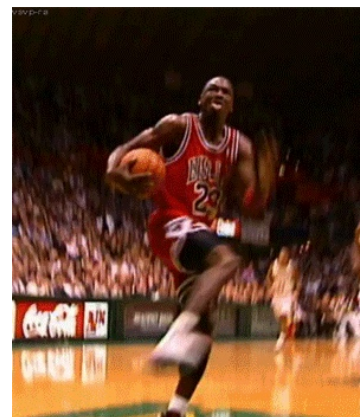
$$x' = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} x, \quad x(0) = x_0.$$

$$e^{At} x_0 = \begin{bmatrix} \cos(t) & -\sin(t) \\ \sin(t) & \cos(t) \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} \quad e^{A \cdot 0} x_0 = \begin{bmatrix} \overset{1}{\cancel{\cos(0)}} & \overset{0}{\cancel{-\sin(0)}} \\ \underset{0}{\cancel{\sin(0)}} & \underset{1}{\cancel{\cos(0)}} \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = x_0.$$

$$x(t) = \begin{bmatrix} x_0 \cos(t) - y_0 \sin(t) \\ x_0 \sin(t) + y_0 \cos(t) \end{bmatrix} = \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}$$

$$x'(t) = \begin{bmatrix} -x_0 \sin(t) - y_0 \cos(t) \\ x_0 \cos(t) - y_0 \sin(t) \end{bmatrix} = \begin{bmatrix} -(x_0 \sin(t) + y_0 \cos(t)) \\ x_0 \cos(t) - y_0 \sin(t) \end{bmatrix} = \begin{bmatrix} -y(t) \\ x(t) \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} x(t) \\ y(t) \end{bmatrix}. \quad \checkmark$$



For the sake of sanity, any better way to compute e^{At} ?

General idea: now that Jordan is out of box, this is easier.
 $AP = PS$.

$$A = PJP^{-1} \quad \text{Jordan block} = \text{"something nice"}$$

$$\begin{aligned} \sum_{n=0}^{\infty} \frac{A^n}{n!} &= \sum_{n=0}^{\infty} \frac{(PJP^{-1})^n}{n!} = \sum_{n=0}^{\infty} \frac{\overbrace{(PJP^{-1})(PJP^{-1}) \dots (PJP^{-1})}^{n \text{ times}}}{n!} \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(P \underbrace{J(P^{-1}P)}_I \underbrace{J(P^{-1}P)}_I \dots \underbrace{J(P^{-1}P)}_I P^{-1} \right) \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(P \underbrace{JJ \dots J}_{n \text{ times}} P^{-1} \right) = P \left(\sum_{n=0}^{\infty} \frac{J^n}{n!} \right) P^{-1} = P e^J P^{-1} \end{aligned}$$

$$J = \begin{bmatrix} a & \\ & b \end{bmatrix} \Rightarrow e^J = \begin{bmatrix} e^a & 0 \\ 0 & e^b \end{bmatrix} \quad \lambda_1 \neq \lambda_2.$$

$$A = \begin{bmatrix} v_1 & v_2 \end{bmatrix} \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix} \begin{bmatrix} v_1 & v_2 \end{bmatrix}^{-1}$$

$$e^{At} x_0 = \underbrace{\begin{bmatrix} v_1 & v_2 \end{bmatrix}}_P \begin{bmatrix} e^{\lambda_1 t} \\ e^{\lambda_2 t} \end{bmatrix} \underbrace{P^{-1} x_0}_{\begin{bmatrix} c_1 \\ c_2 \end{bmatrix}} = \underbrace{c_1 v_1 e^{\lambda_1 t}}_{\text{straight line sol.}} + \underbrace{c_2 v_2 e^{\lambda_2 t}}_{\text{straight line sol.}}$$

7:49

Step-by-step. Consider the initial value problem

$$X' = AX \quad X(0) = X_0$$

a) $A = \begin{bmatrix} -3 & 0 \\ 0 & 2 \end{bmatrix}$, $X_0 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Step 0: eigenvalues :

a) 1, 6

b) 2, 3

c) $\overset{\lambda_2}{2}, \overset{\lambda_1}{-3}$ pol 1.

d) -1, 6

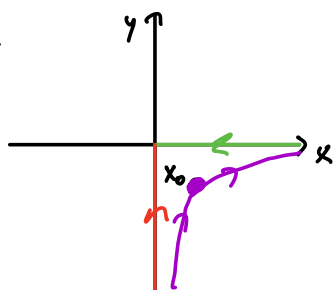
eigenvectors : $v_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ $v_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Step 1: Compute $e^{At} = P \begin{bmatrix} e^{-3t} & 0 \\ 0 & e^{2t} \end{bmatrix} P^{-1} = \begin{bmatrix} e^{-3t} & 0 \\ 0 & e^{2t} \end{bmatrix}$
 $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = [v_1 \ v_2]$ $I^{-1} = I$

Step 2: State your solution

$$X(t) = e^{At} \cdot X_0 = \begin{bmatrix} e^{-3t} & 0 \\ 0 & e^{2t} \end{bmatrix} \begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} x_0 e^{-3t} \\ y_0 e^{2t} \end{bmatrix} = \begin{bmatrix} e^{-3t} \\ -e^{2t} \end{bmatrix}$$

Extra Step: Sketch it
For now



$$(e^{-3t}, -e^{2t})$$
$$(x, -x^{-2/3}) = (x, -\frac{1}{x^{2/3}})$$

$$b) X = \begin{bmatrix} -3 & -5 \\ 0 & 2 \end{bmatrix}, X_0 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -3 & -5 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -3+5 \\ -2 \end{bmatrix} = \begin{bmatrix} 2 \\ -2 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Step 0: Eigenvalues: $-3, 2$

Eigenvectors:

$$a) \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$c) \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$b) \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$d) \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

pol 2.

Step 1.: Compute e^{At}

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \cdot \frac{1}{ad-bc}$$

$$\begin{aligned} e^{At} &= \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \cdot \begin{bmatrix} e^{-3t} & 0 \\ 0 & e^{2t} \end{bmatrix} \cdot \begin{bmatrix} +1 & +1 \\ 0 & -1 \end{bmatrix} \cdot \frac{1}{\cancel{1-0}} \\ &= \begin{bmatrix} e^{-3t} & e^{2t} \\ 0 & -e^{2t} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \\ &= \begin{bmatrix} e^{-3t} & e^{-3t} - e^{2t} \\ 0 & e^{2t} \end{bmatrix} \end{aligned}$$

Step 2:

$$\begin{aligned}
 x(t) = e^{At} x_0 &= \begin{bmatrix} e^{-3t} & e^{-3t} - e^{2t} \\ 0 & e^{2t} \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\
 &= \begin{bmatrix} e^{-3t} - e^{2t} \\ e^{2t} \end{bmatrix} = e^{-3t} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + e^{2t} \begin{bmatrix} -1 \\ 1 \end{bmatrix}
 \end{aligned}$$

"Extra"-step. Sketch.

$t \sim -\infty$

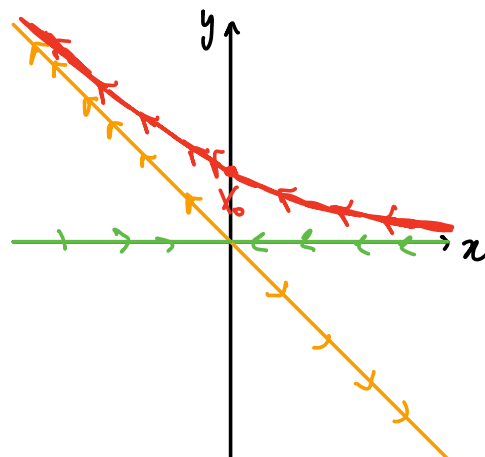
$\text{orange} \approx 0$

$\text{red} \approx \text{green}$

$t \rightarrow \infty$

$\text{green} \approx 0$

$\text{red} \approx \text{orange}$



~~lecture 9~~

What does it mean in the grand scheme of things?

The Trinity



2 distinct real eigenvalues

$$\textcircled{\text{I}} \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix}$$

$\underbrace{\hspace{1.5cm}}_{\mathcal{J}}$

$$\textcircled{\text{II}} \begin{bmatrix} \lambda & \\ & \bar{\lambda} \end{bmatrix}$$

2 distinct complex eigenvalues

$$\textcircled{\text{III}} \begin{bmatrix} \lambda & 1 \\ & \lambda \end{bmatrix}$$

Repeated eigenvalues.

Natural Question:

Q: Why is enough to consider only those?

Answer: Suppose $x' = Ax$ for some $A = P\mathcal{J}P^{-1}$.

$y = P^{-1}x$

Consider the change of variables $y = P^{-1}x = \begin{bmatrix} u \\ v \end{bmatrix}$

$$y' = (P^{-1}x)' = P^{-1}(x') = P^{-1}(Ax) = P^{-1}(P\mathcal{J}P^{-1})x = \underbrace{(P^{-1}P)}_{\text{I}} \underbrace{\mathcal{J}}_Y (P^{-1}x)$$

$$y_0 = P^{-1}x_0$$

$$y' = \mathcal{J}y \Rightarrow y(t) = e^{\mathcal{J}t} y_0$$

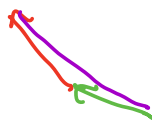
$$y = P^{-1}x \Rightarrow x = Py$$

$$x(t) = P(e^{\mathcal{J}t} y_0) = \underbrace{Pe^{\mathcal{J}t}P^{-1}}_{e^{At}} x_0 = e^{At} x_0$$

It is ENOUGH to study $y' = \mathcal{J}y$.



qualitative
~
~



3.3 Real Eigenvalues.

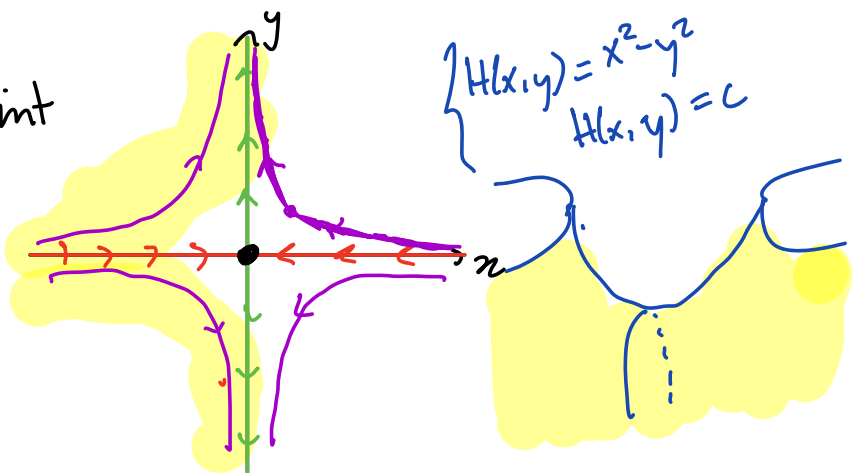
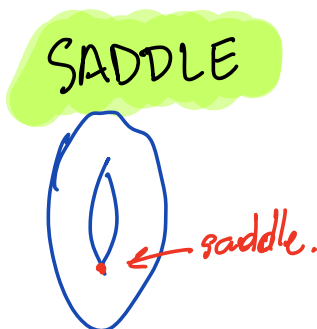
Suppose $A = \begin{bmatrix} \lambda_1 & \\ & \lambda_2 \end{bmatrix}$,

λ_1, λ_2
distinct non-zero
real eigenvalues.

i.e., $x' = \lambda_1 x \Rightarrow c_1 \cdot e^{\lambda_1 t}$
 $y' = \lambda_2 y \Rightarrow c_2 \cdot e^{\lambda_2 t}$

$c_1 \underbrace{\begin{bmatrix} 1 \\ 0 \end{bmatrix}}_{v_1} e^{\lambda_1 t} + c_2 \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{v_2} e^{\lambda_2 t}$

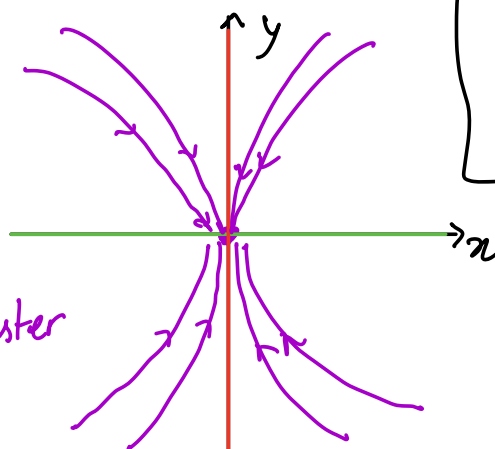
if $\lambda_1 < 0 < \lambda_2$
(0,0) is equilibrium point



if $\lambda_1 < \lambda_2 < 0$

$e^{\lambda_1 t} \rightarrow 0$
 $(e^{\lambda_1 t}, e^{\lambda_2 t}) \xrightarrow{t \rightarrow \infty} (0,0)$

$e^{\lambda_1 t}, e^{\lambda_2 t}$
 $e^{-\lambda_1 t}, e^{-\lambda_2 t}$
 x decreases faster



$t \rightarrow \infty$
 $\|x\| \approx 0$ faster
 ~~$\|x\| \approx 0$~~

SINK
stable point

if $0 < \lambda_2 < \lambda_1$

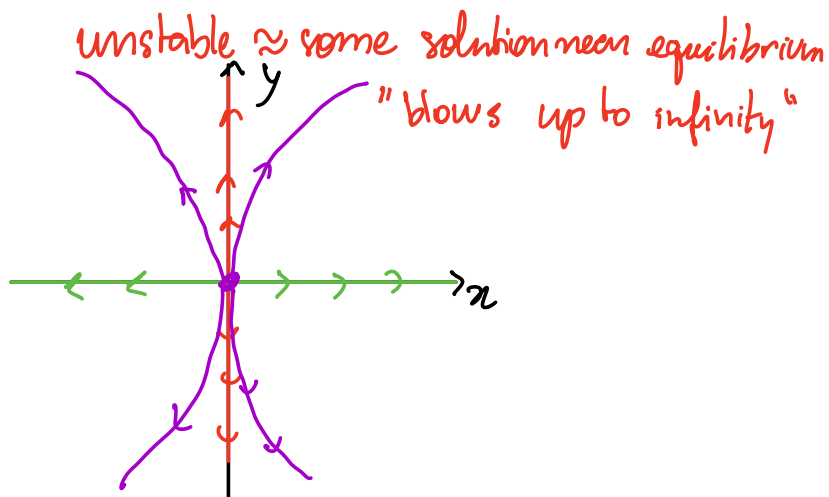
SOURCE

unstable

$t \rightarrow -\infty$

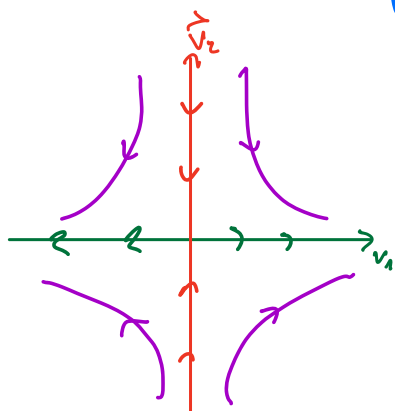
$m \approx m$

$m \approx 0$



In general, $x' = Ax$, λ_1, λ_2 real eigenvalues, v_1, v_2 eigenvect.

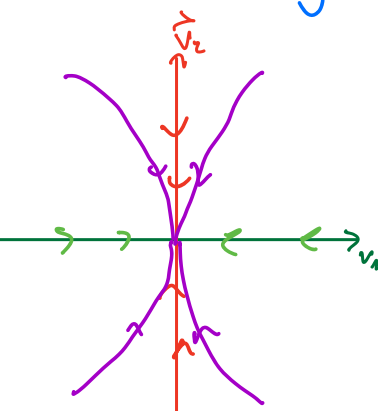
think about: (qualitative $\approx$ general picture near equilibrium)



$\lambda_1 < 0 < \lambda_2$

saddle

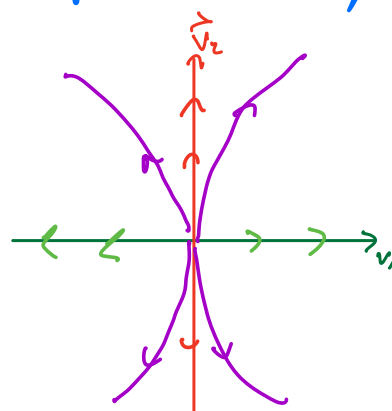
(unstable)



$\lambda_1 < \lambda_2 < 0$

sink

(stable)



$0 < \lambda_2 < \lambda_1$

source

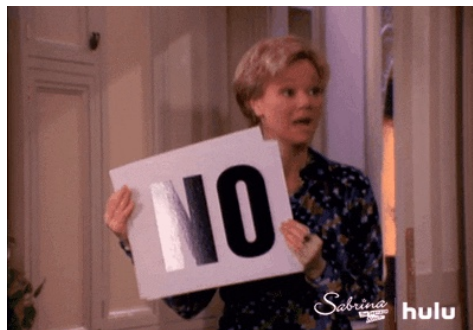
(unstable)

3.4. Complex Eigenvalues.

Suppose $A = \begin{bmatrix} a & b \\ -b & a \end{bmatrix}$ with eigenvalues $\lambda = a \pm bi$

Check those are the eigenvalues.

Are we going to find straight lines here?



Great video about Complex numbers:

Youtube: Veritasium, "How imaginary numbers were invented."

Anyway...

the same reasoning applies:

Suppose $\lambda = a + bi$ is the eigenvalue, v eigenvector.

then

$$X(t) = c_1 v e^{\lambda t} + c_2 \bar{v} e^{\bar{\lambda} t}$$

is a complex-valued solution.

but :

$$e^{\lambda t} = e^{(a+bi)t} = e^a (\cos(bt) + i \sin(bt)) = e^a \cos(bt) + i e^a \sin(bt)$$

What about the eigenvectors?

$$\lambda = a + bi$$

$$[A - (a+bi)I] \begin{bmatrix} x \\ y \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} -bi & b \\ -b & -bi \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = 0 \Rightarrow \begin{aligned} -bi x + by &= 0 \\ y &= ix \end{aligned}$$

$$v = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

What is $ve^{\lambda t}$?

$$\begin{bmatrix} 1 \\ i \end{bmatrix} e^{(a+bi)t} = \begin{bmatrix} e^a \cos(bt) + i e^a \sin(bt) \\ -e^a \sin(bt) + i e^a \cos(bt) \end{bmatrix}$$

$$= e^a \begin{bmatrix} \cos(bt) \\ -\sin(bt) \end{bmatrix} + i e^a \begin{bmatrix} \sin(bt) \\ \cos(bt) \end{bmatrix}$$

$x_1(t)$

$$\text{Re}(ve^{\lambda t}) = \frac{ve^{\lambda t} + \bar{v}e^{\bar{\lambda}t}}{2}$$

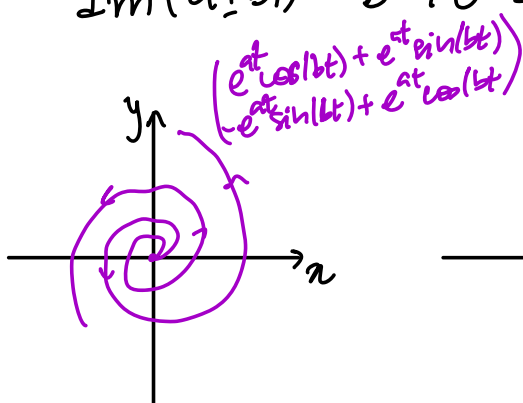
$$\text{Im}(ve^{\lambda t}) = \frac{ve^{\lambda t} - \bar{v}e^{\bar{\lambda}t}}{2}$$

$x_2(t)$.

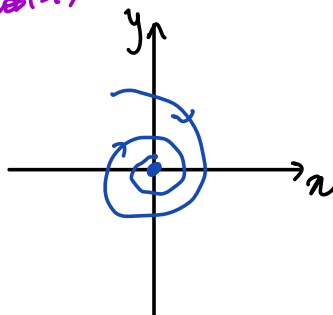
What kind of nice pictures can we have?

$$\operatorname{Re}(a \pm bi) = \begin{cases} a > 0 \Rightarrow \text{unstable} & (e^{at} \rightarrow \infty \text{ as } t \rightarrow \infty) \\ a < 0 \Rightarrow \text{stable} & (e^{at} \rightarrow 0 \text{ as } t \rightarrow \infty) \\ a = 0 \Rightarrow \text{---} & e^{at} = 1. \end{cases}$$

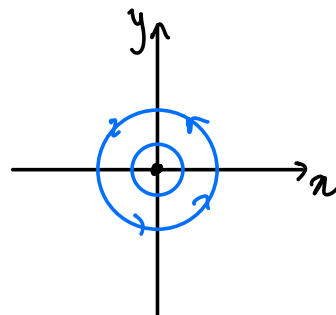
$\operatorname{Im}(a \pm bi) = b \neq 0 \Rightarrow \text{rotation}$ (I mean, $\mathbb{C}$ needs $b \neq 0$)



$a > 0, b \neq 0$
SPIRAL SOURCE



$a < 0, b \neq 0$
SPIRAL SINK



$a = 0, b \neq 0$.
CENTER.
 $x' = Ax$

— the end. Classification of type of equilibria.